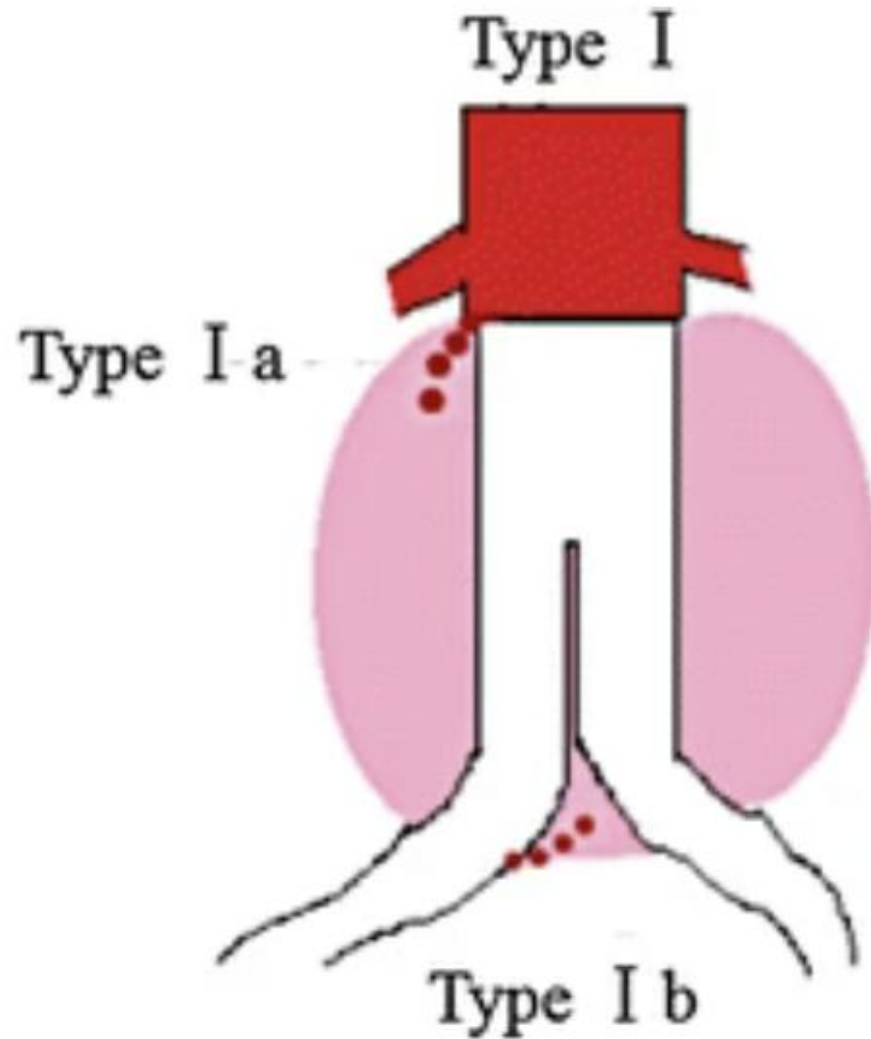


KELOS Endovascular Stent

CHIH-HAN (MATTHEW) CHEN, RYAN POTECHIN, MARK ROTERING,
FAITH TONER, & MEGAN WIRTJES



Unmet Clinical Need

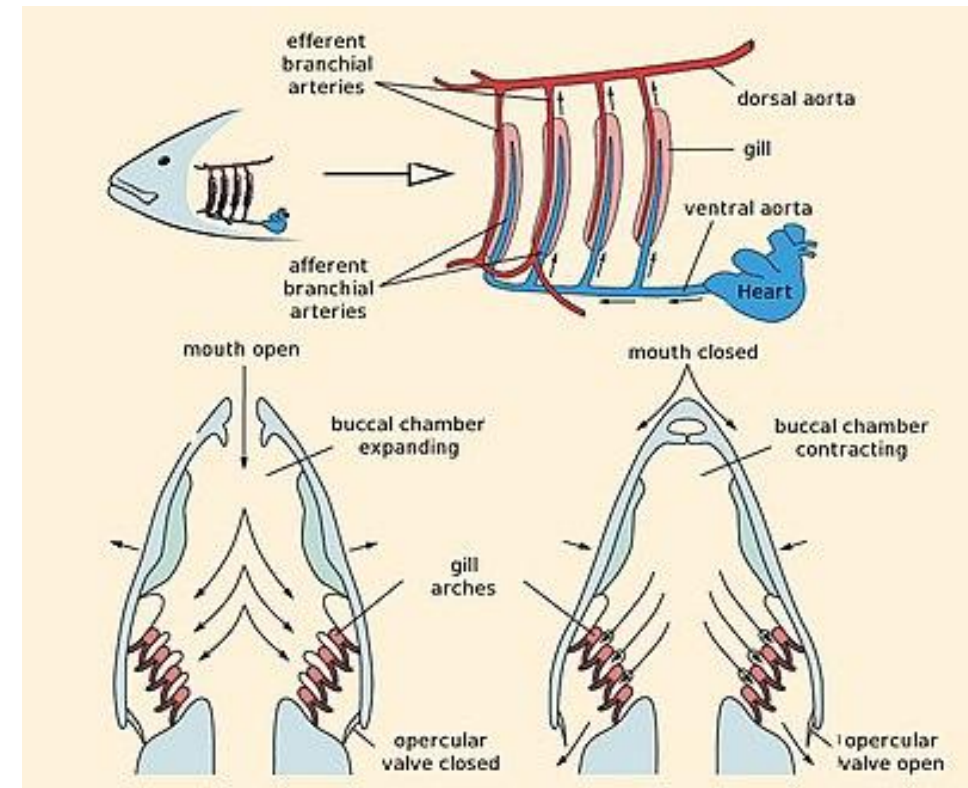
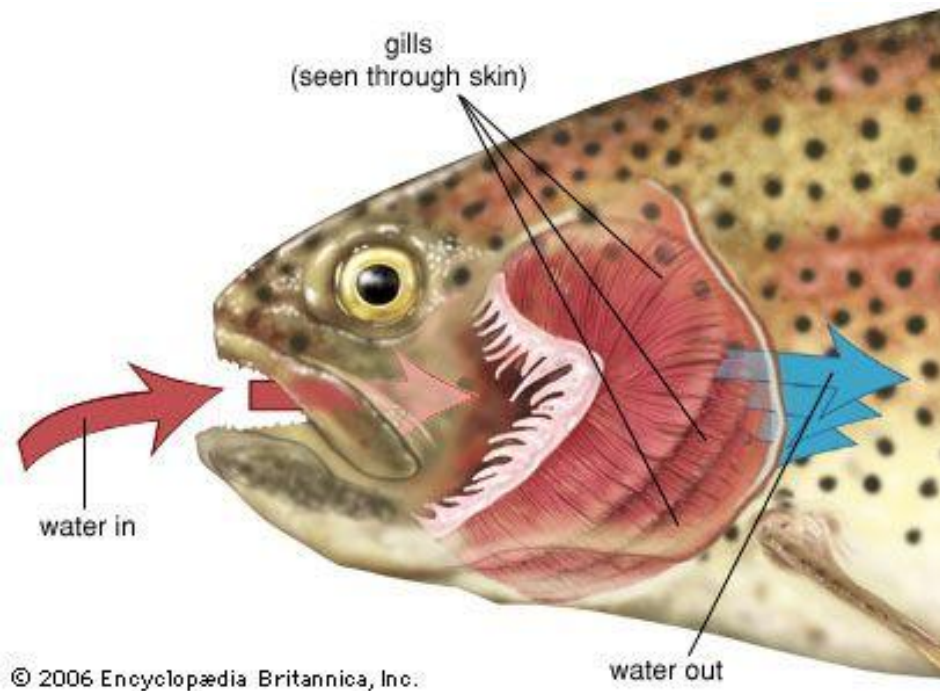
- Abdominal Aortic Aneurysm (AAA)
- Endovascular Aneurysm Repair (EVAR)
- Type 1a endoleaks
 - Most likely to cause rupture
 - Difficult to manage and prevent

"Abdominal aortic aneurysm causes, symptoms, treatments," Causes, Symptoms, Treatments - UChicago Medicine, <https://www.uchicagomedicine.org/conditions-services/heart-vascular/aortic-disease/abdominal-aortic-aneurysm> (accessed Mar. 17, 2024).

"Endovascular stent graft repair," UNMC, <https://www.unmc.edu/surgery/divisions/vascular/patient-care/procedures/endovascular-stent.html> (accessed Mar. 17, 2024).

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Design Inspiration



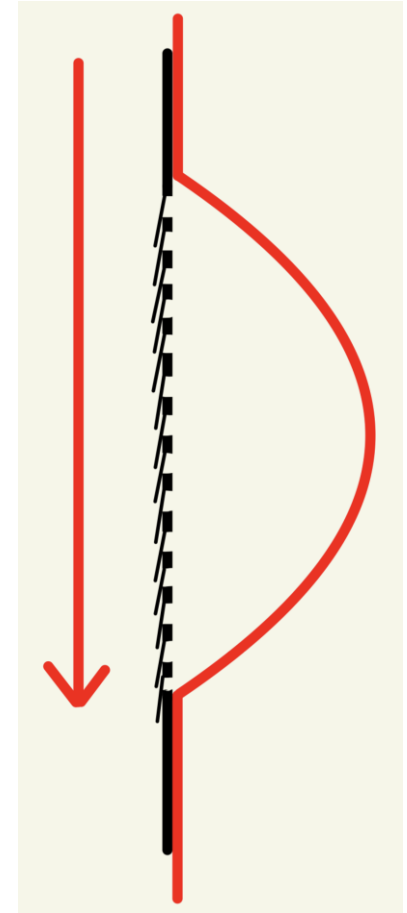
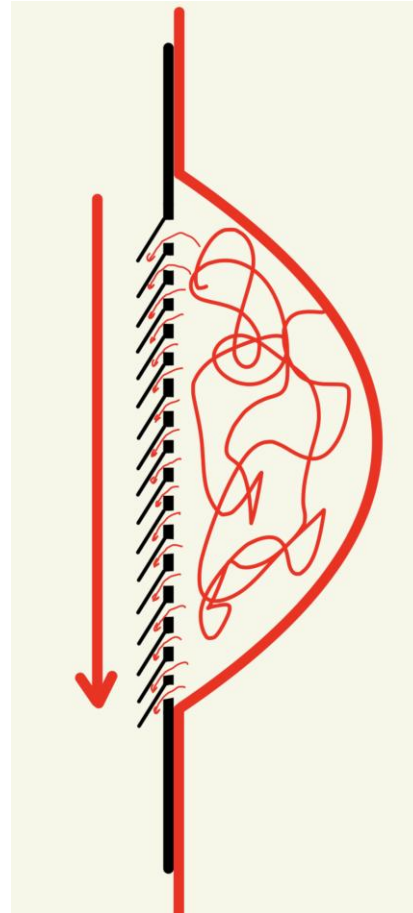
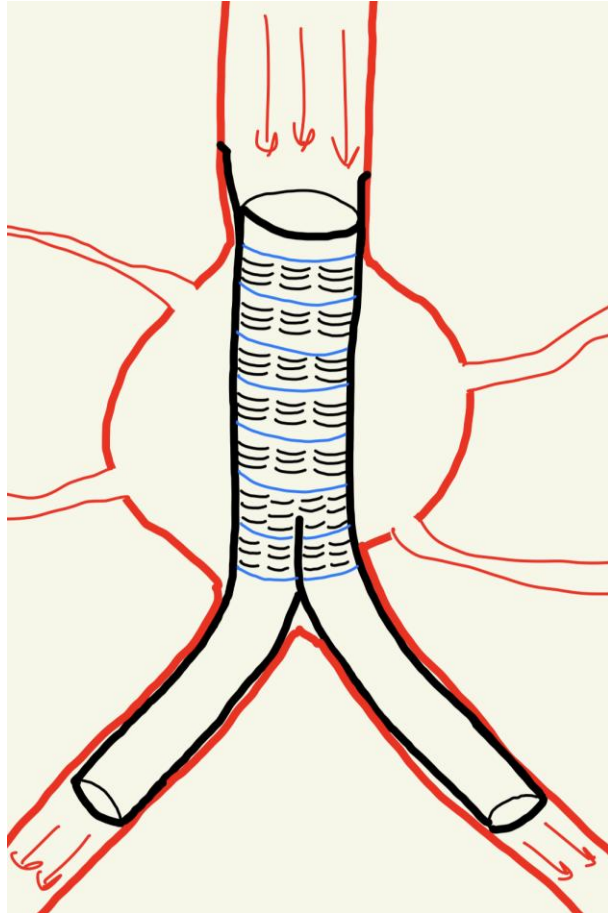
[1] P. D. of F. Richard J. Strange, WFS 550 fish physiology- respiratory pump, <https://web.utk.edu/~rstrange/wfs550/html-con-pages/f-respiratory-pump.html#:~:text=In%20most%20fish%2C%20the%20system,the%20gill%20surface%20as%20possible>. (accessed Mar. 17, 2024).

[2] P. D. of F. Richard J. Strange, WFS 550 Fish Physiology - Vascular System, <https://web.utk.edu/~rstrange/wfs550/html-con-pages/m-vascular-system.html> (accessed Mar. 17, 2024).

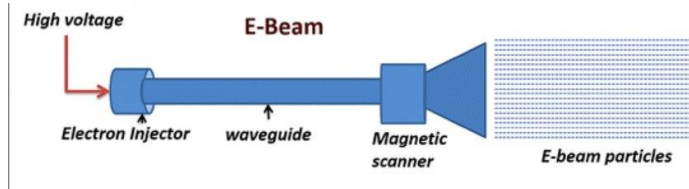
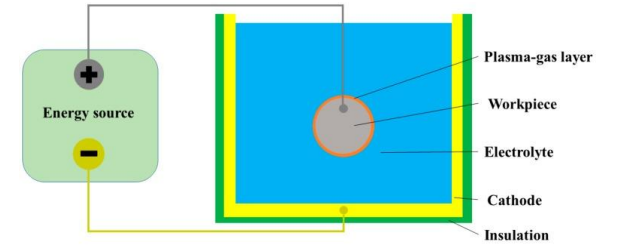
[3] J. M. Wilson and P. Laurent, "Fish gill morphology: Inside out," *Journal of Experimental Zoology*, vol. 293, no. 3, pp. 192–213, Jun. 2002. doi:10.1002/jez.10124

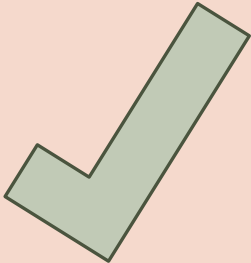
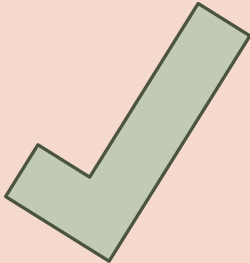
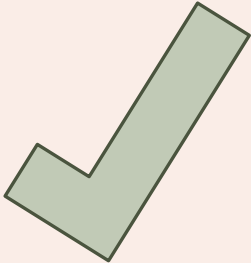
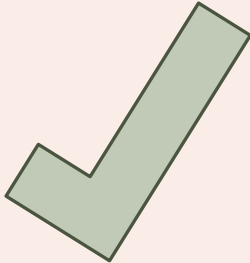
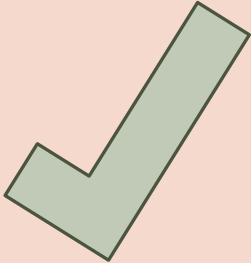
[4] "The respiratory system," Encyclopædia Britannica, <https://www.britannica.com/animal/fish/The-respiratory-system> (accessed Mar. 17, 2024).

KELOS Design



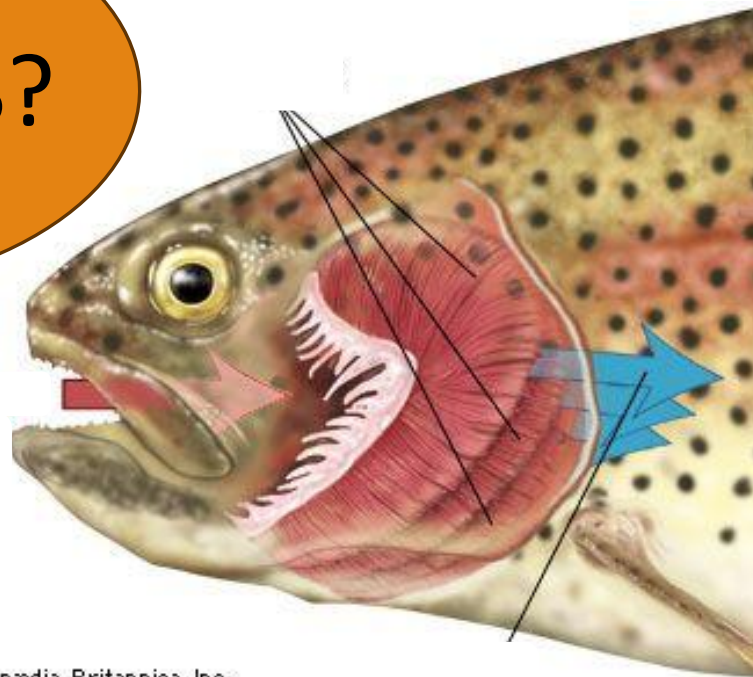
Cleaning & Sterilization



	KELOS Endovascular Stent	Gore® EXCLUDER® AAA ENDOPROSTHESIS	ALTO™ Abdominal Stent Graft System
Modular Sealing Mechanism			
Separation of Fixation and Sealing			
Adaptable to Prevent Rupture			

Biomimetics

QUESTIONS?



© 2006 Encyclopædia Britannica, Inc.

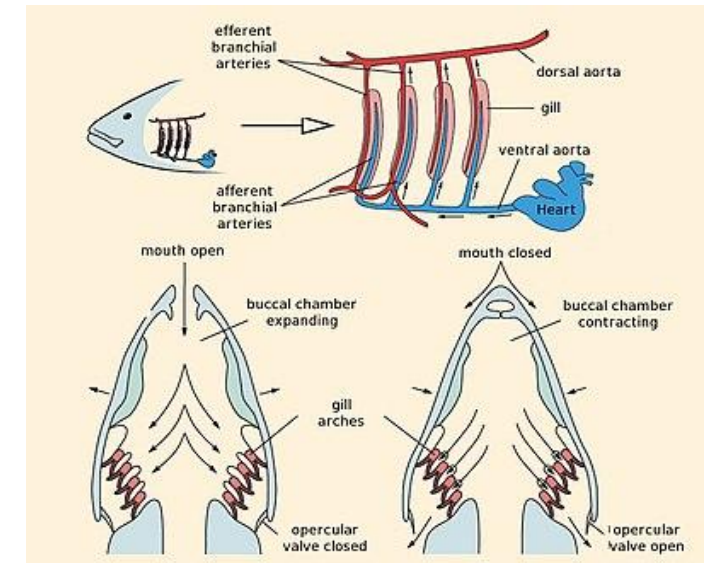
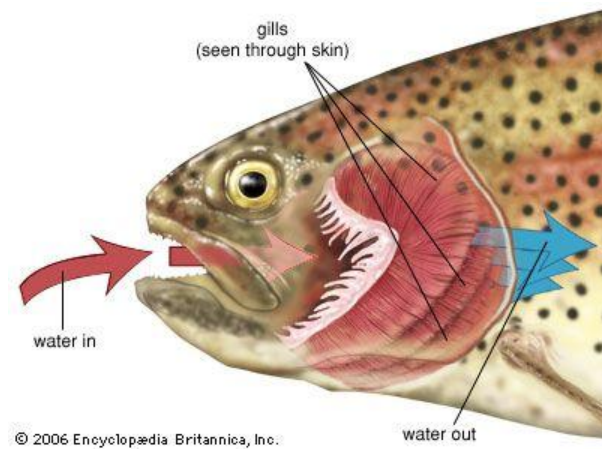
A solid orange vertical bar on the left side of the slide.

Design

Inspiration from Nature

To breathe fish pass water in through their mouth and over and out through their gills

- In this process the gills on the fish act as a valve
- Allow for one directional flow from the body of the fish to the water surrounding the fish
- Ensure that there is no backflow from surrounding water into the body of the fish



[1] P. D. of F. Richard J. Strange, WFS 550 fish physiology- respiratory pump, <https://web.utk.edu/~rstrange/wfs550/html-con-pages/f-respiratory-pump.html#:~:text=In%20most%20fish%2C%20the%20system,the%20gill%20surface%20as%20possible>. (accessed Mar. 17, 2024).

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[3] J. M. Wilson and P. Laurent, "Fish gill morphology: Inside out," *Journal of Experimental Zoology*, vol. 293, no. 3, pp. 192–213, Jun. 2002. doi:10.1002/jez.10124

[4] "The respiratory system," Encyclopædia Britannica, <https://www.britannica.com/animal/fish/The-respiratory-system> (accessed Mar. 17, 2024).

New Stent Design

Traditional stent design

- Use of Nitinol for structure
- Use of ePTFE for walls of the stent

Inner layer with overlapping leaflets

- Acts as a one directional valve
 - Allows for normal fluid flow through the main vessel
 - Allows blood to enter the main vessel from the sac of the aneurysm
- Prevents fluid flow out of the main vessel
- Gills only open when pressure in the aortic sac is high enough
 - Pressure difference can differ by patient so more research needs to be done



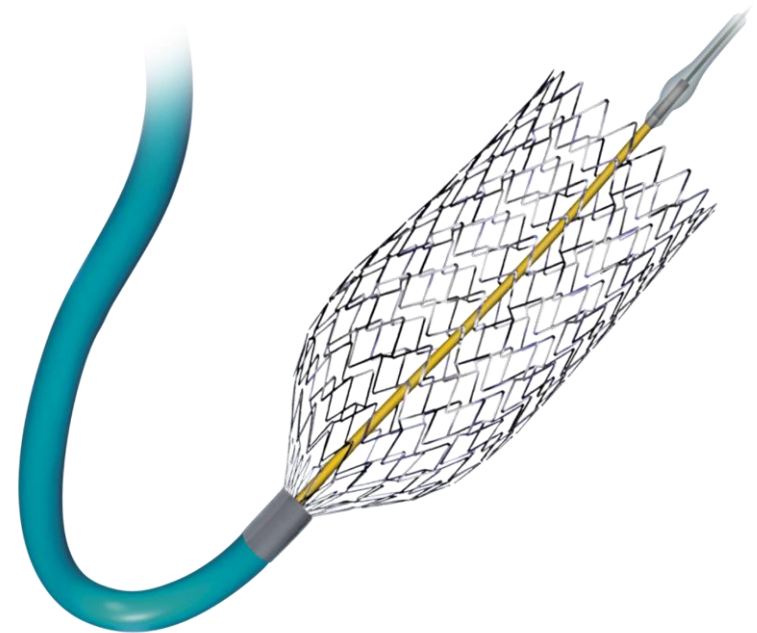
[1] "The respiratory system," Encyclopædia Britannica, <https://www.britannica.com/animal/fish/The-respiratory-system> (accessed Mar. 17, 2024).

Materials

Expanded Polytetrafluoroethylene
(ePTFE)



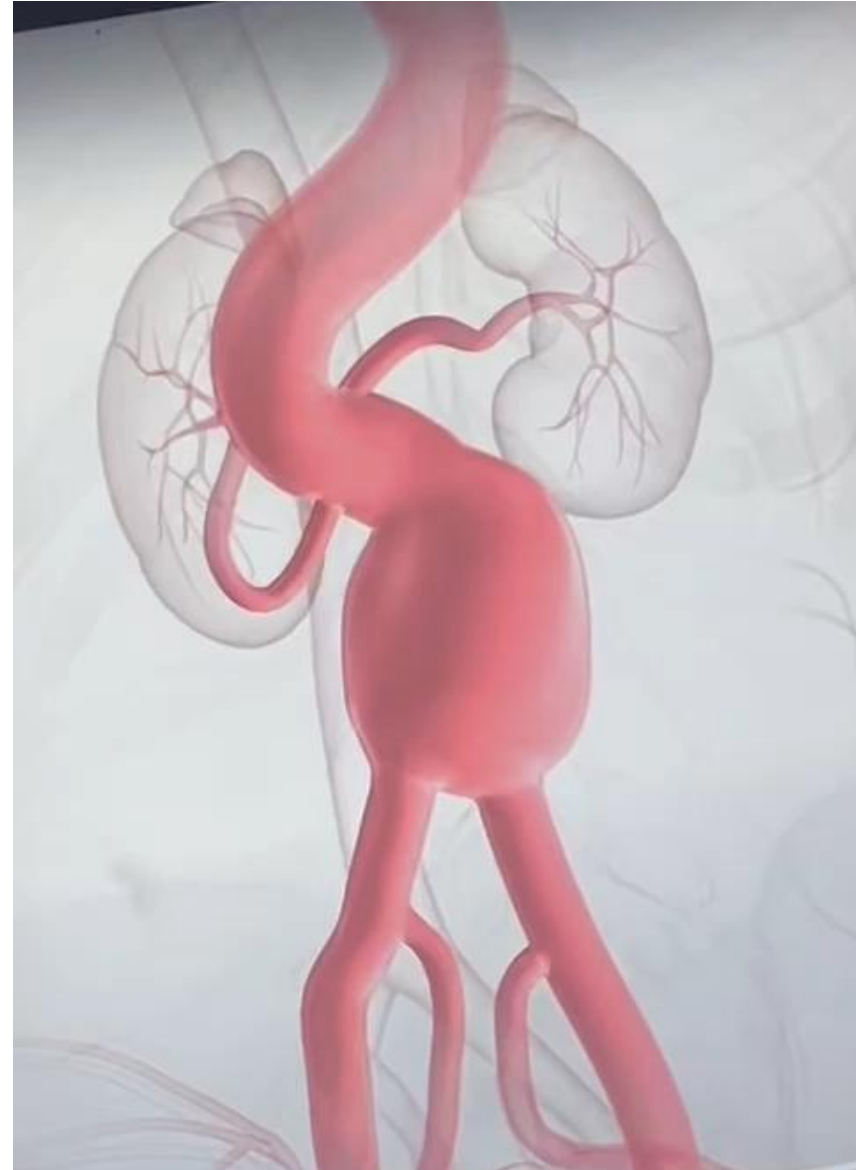
Nitinol (Nickel and Titanium)



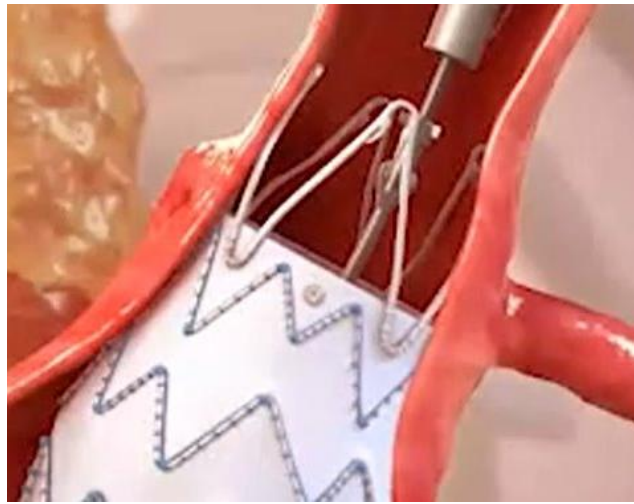
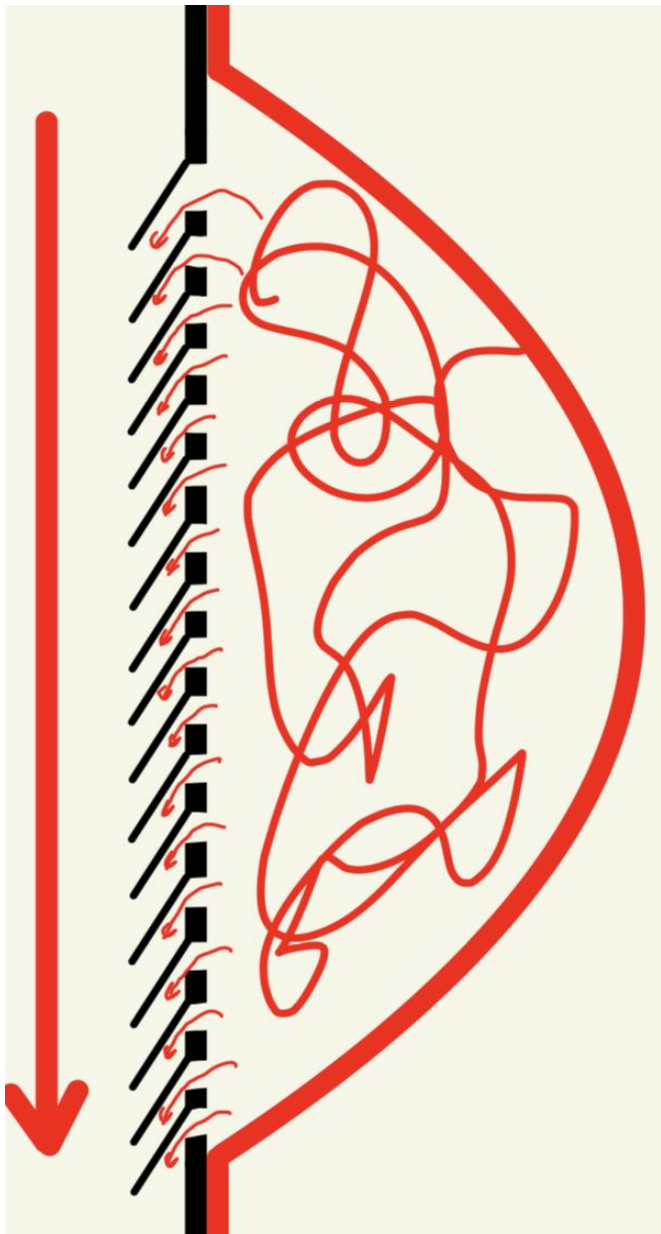
Human Factors & Clinical Procedure

GOAL:

Mitigate risk of aneurysm rupture and prolong defective stent life until a revision surgery can be performed



GORE Stent Used for Example Video



Why its better

Something New :

Gill-like valve fail-safe allowing for unidirectional blood flow from aneurysm sac to stent lumen

Something Borrowed:

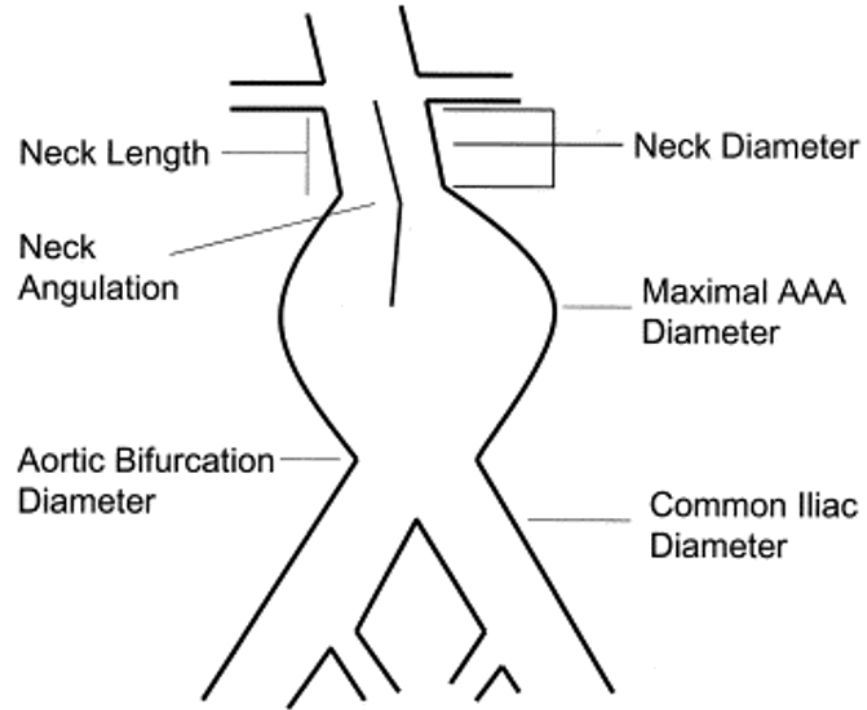
1. Isolation of fixation and sealing
2. Modular sealing mechanism
3. 12F catheter profile
4. Traditional Materials

Potential Questions

-Achilles Heel-

Natural Anatomical and Physiological Variance

ANEURYSM NECK LENGTH



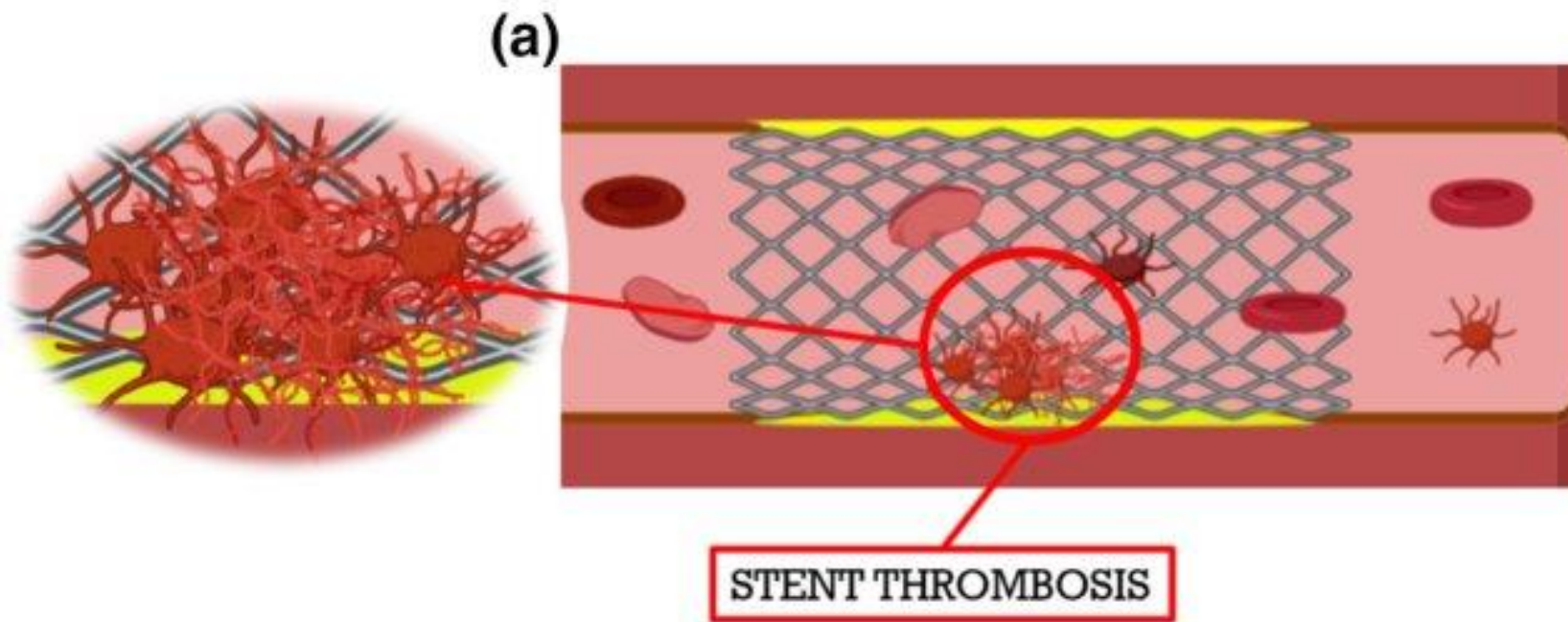
ANEURYSM BURST PRESSURE



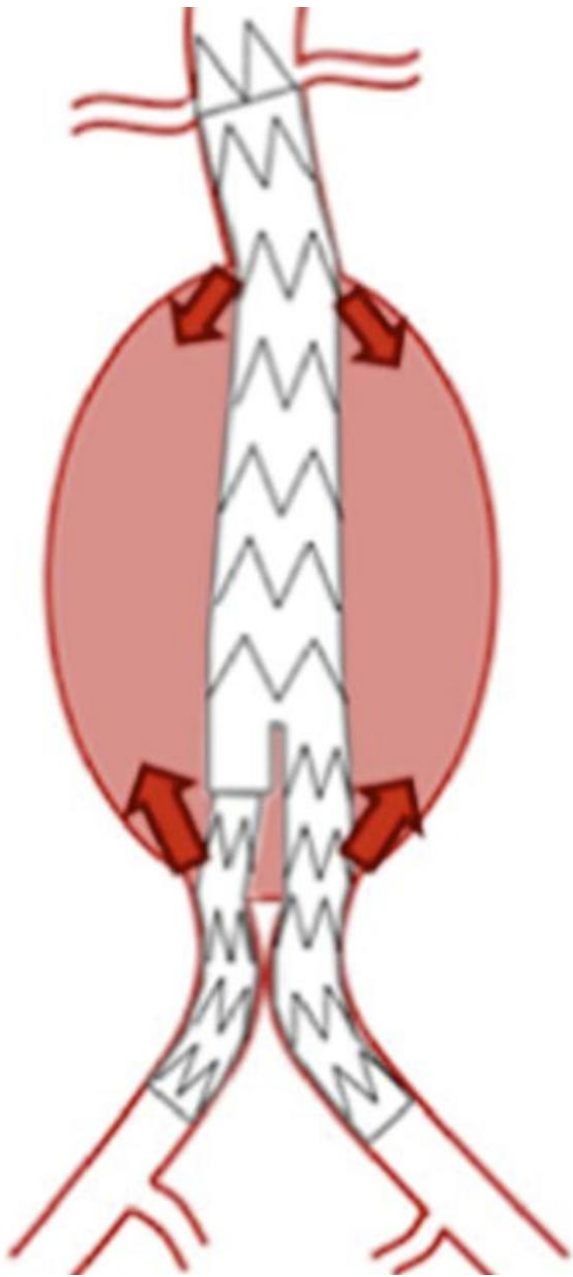
Material & Design



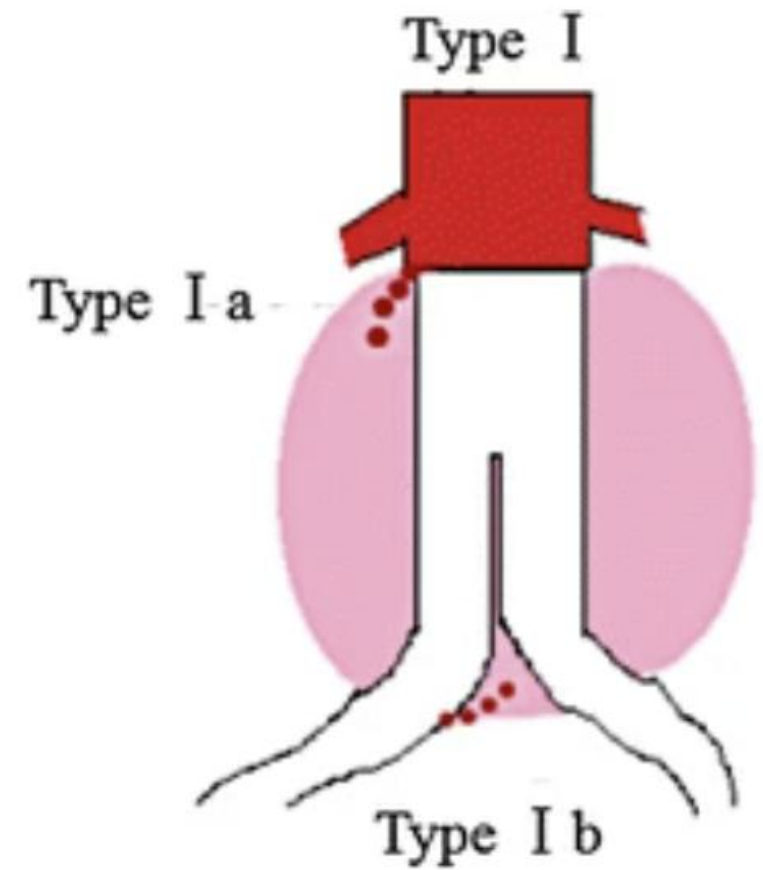
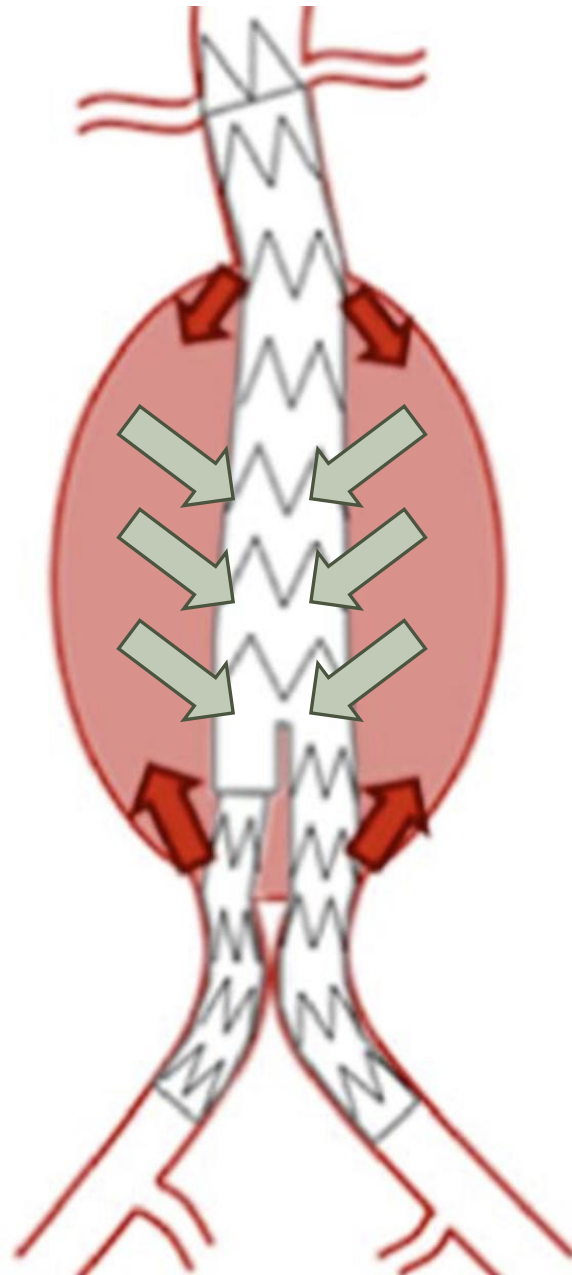
- Material Selection
- Gill Spacing
- Open Pressure & Mechanism
- Gill Fatigue



Biocompatibility



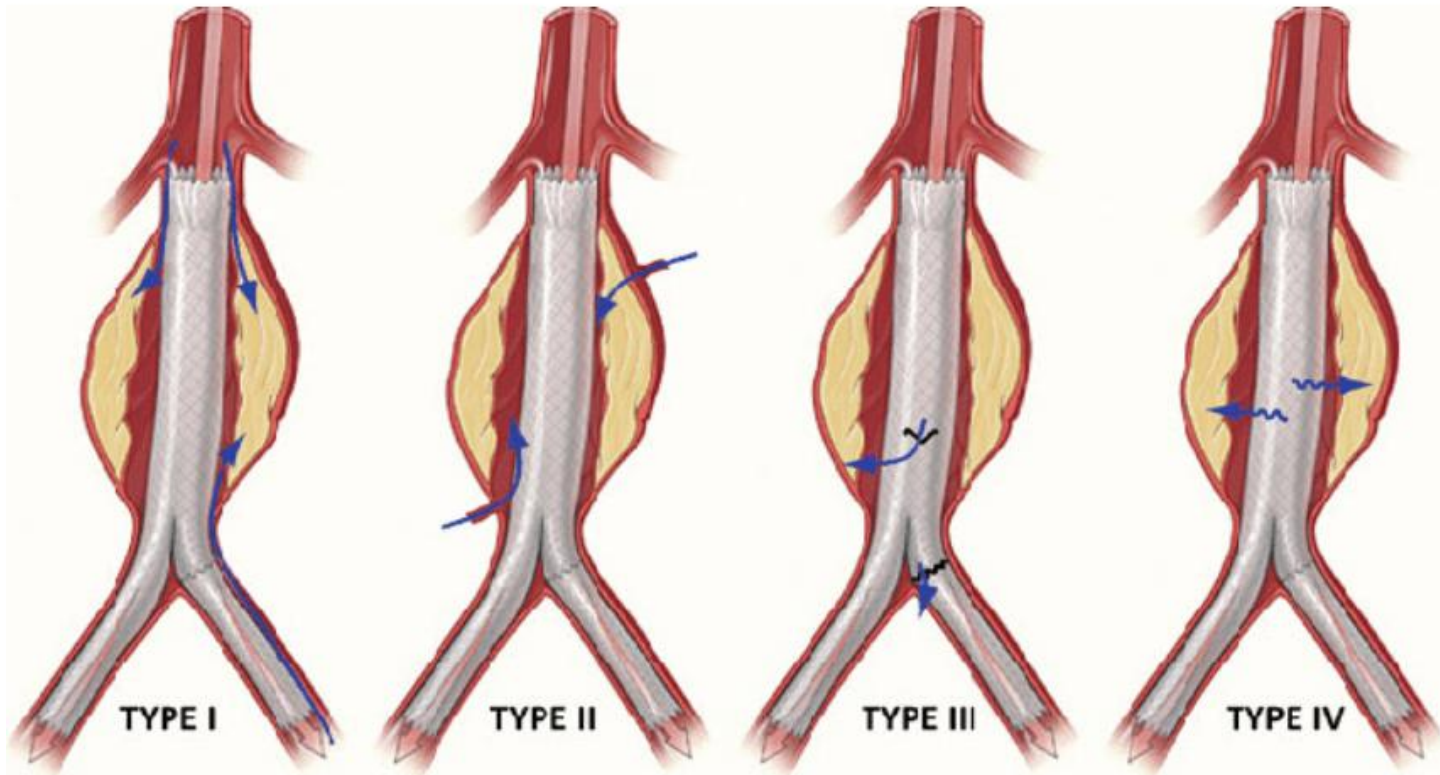
VS



Leak
Notification



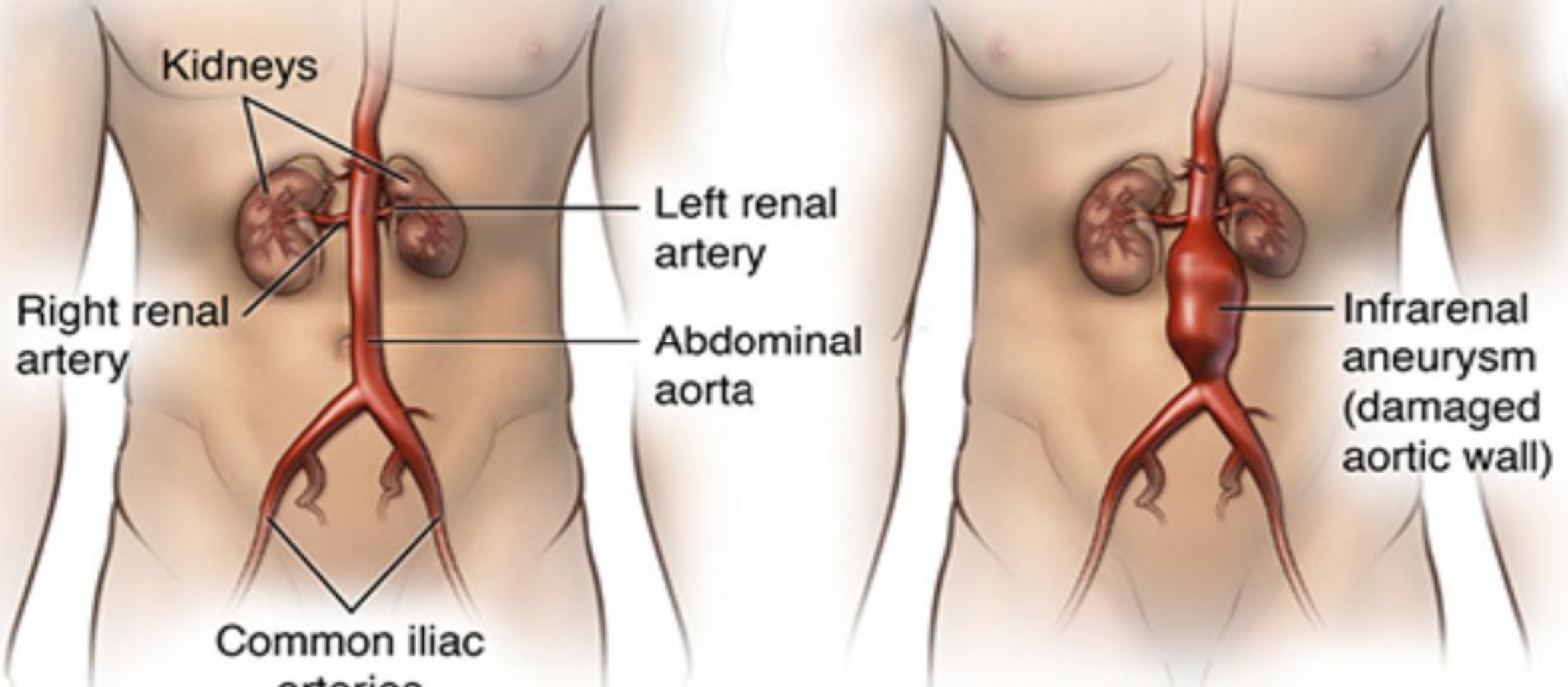
Background



Endovascular Leak Type 1A

The Issue

- AAA Rupture causes 4-5% of sudden deaths in the US
- Aneurysm is a 1.5 diameter increase from original aortic diameter
- One of the most serious medical emergencies
- Approximately 10% of all EVAR case will develop a type 1a endoleak
- Difficult to predict, prevent, and manage



Abdominal Aortic Aneurysm (AAA)

"Abdominal aortic aneurysm causes, symptoms, treatments," Causes, Symptoms, Treatments - UChicago Medicine, <https://www.uchicagomedicine.org/conditions-services/heart-vascular/aortic-disease/abdominal-aortic-aneurysm> (accessed Mar. 17, 2024).

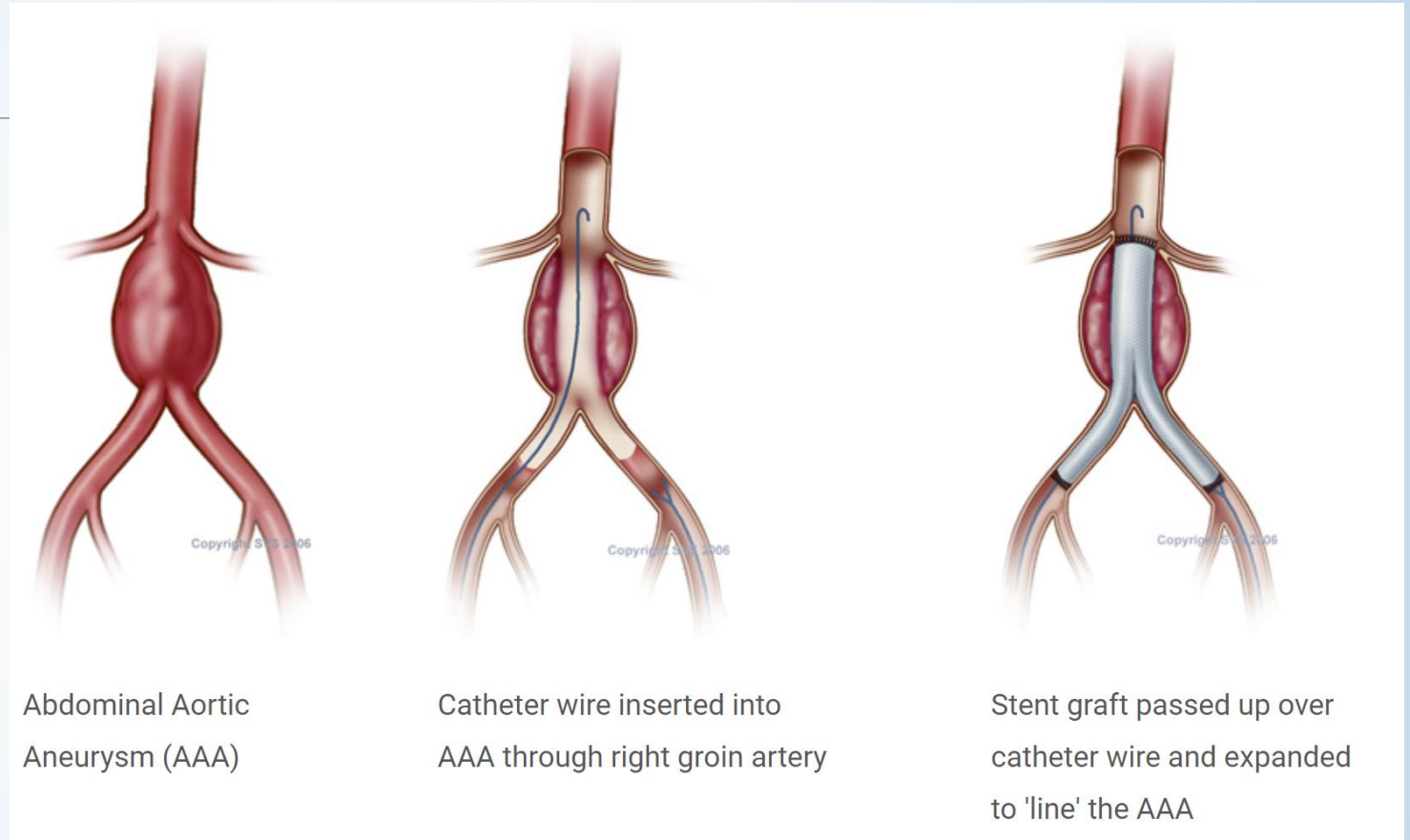
EVAR Procedure

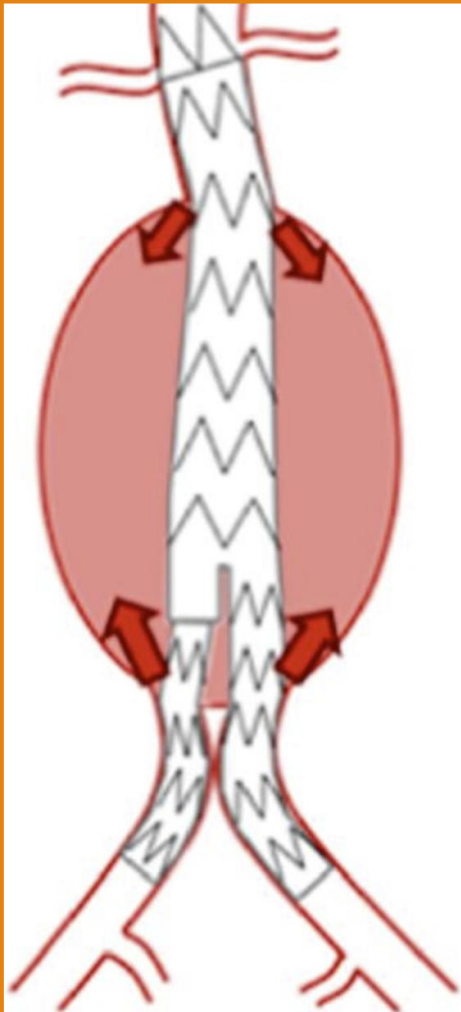
Insertion through femoral artery

X-ray and/or Ultrasound imaging used concurrently for proper placement

Creates seal to artery wall

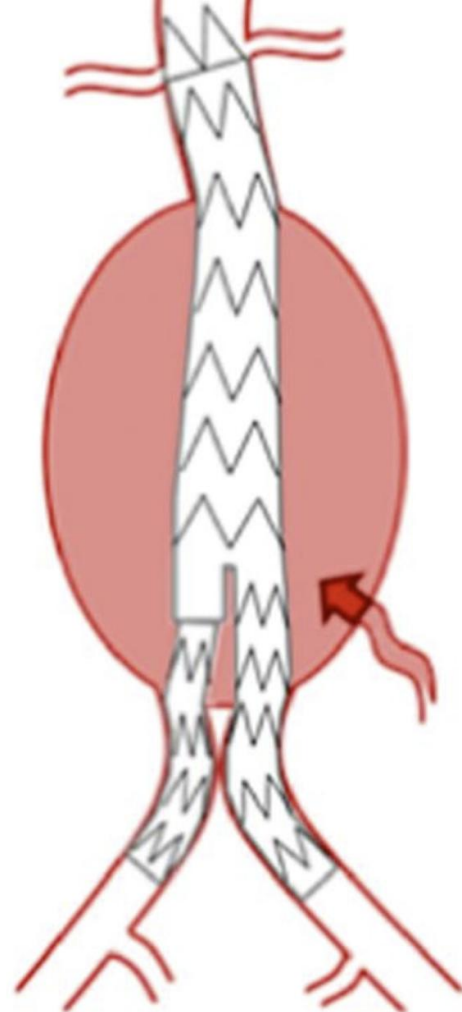
Keeps the aneurysm from expanding further





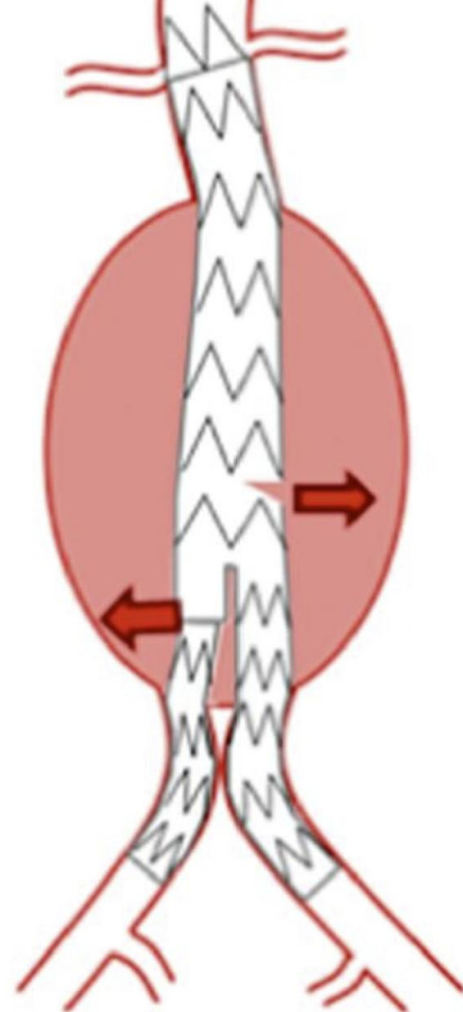
TYPE I

Proximal or distal graft attachment site leaks.



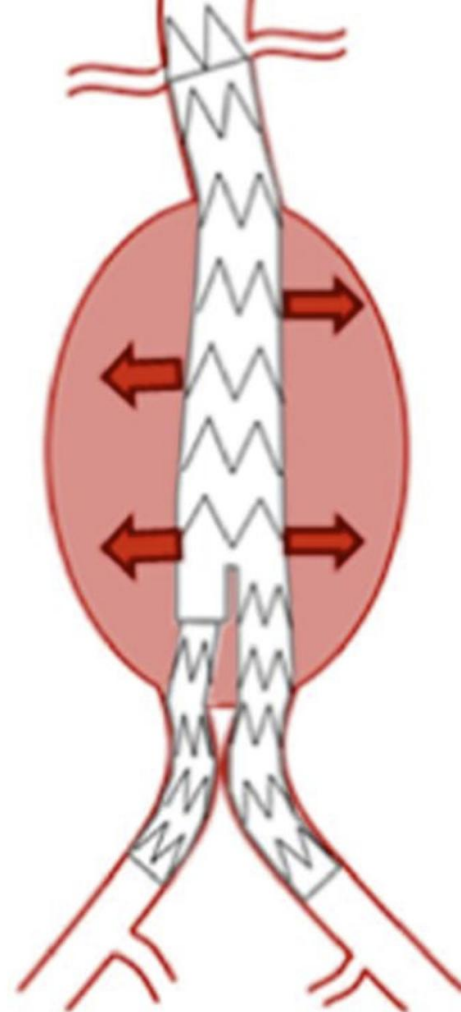
TYPE II

Retrograde flow into the aneurysm sac from aortic side-branches such as the lumbar or inferior mesenteric arteries.



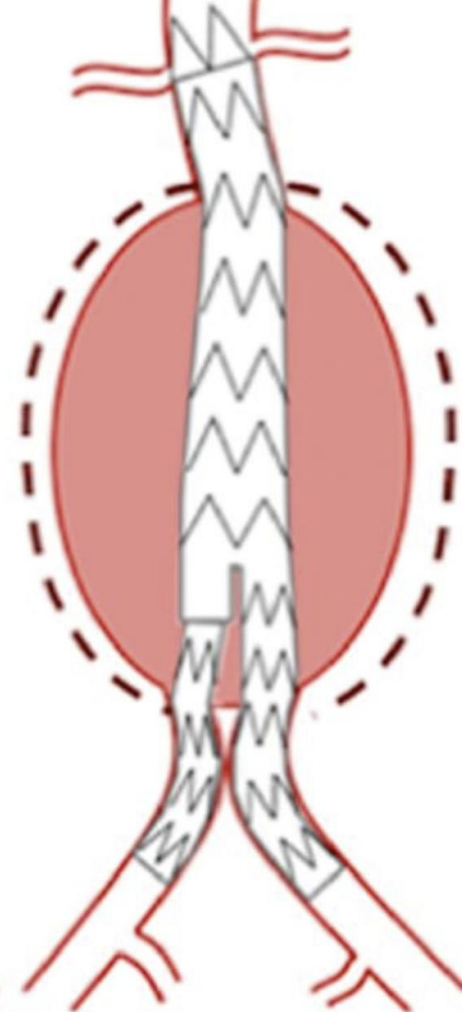
TYPE III

Caused by a defect in the graft either due to a fabric tear or disconnection of modular overlap.



TYPE IV

Graft wall porosity.



TYPE V

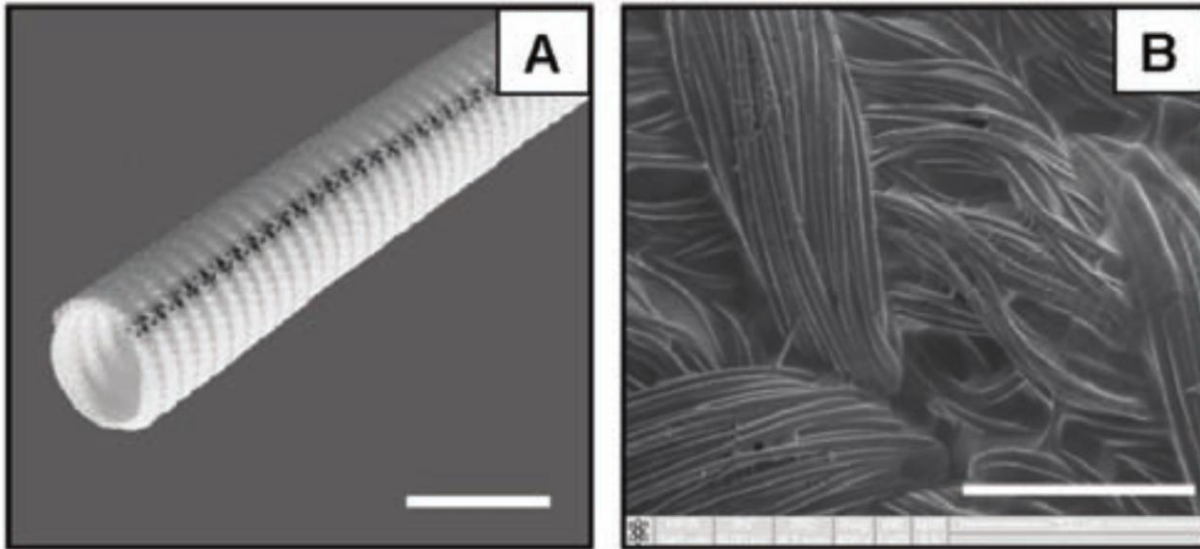
Increase in maximum aneurysm diameter with no identifiable endoleaks.

Tissue Responses

Thrombosis		Restenosis	
Pathophysiology	Considerations	Pathophysiology	Considerations
○ Under expanded stent or small vessels – stasis and turbulence ○ Delayed healing or edge dissection – endothelial injury or disruption	○ Antiplatelet Therapy (Anticoagulation) ○ Thrombolytic Therapy	○ Scar tissue forming under stent lining ○ Neointimal Hyperplasia ○ Migration of smooth muscle cells resulting in thickening of vascular wall	○ Medication ○ Drug-Eluting Stent

Potential Graft Materials

Polyethylene Terephthalate (PET/Dacron)



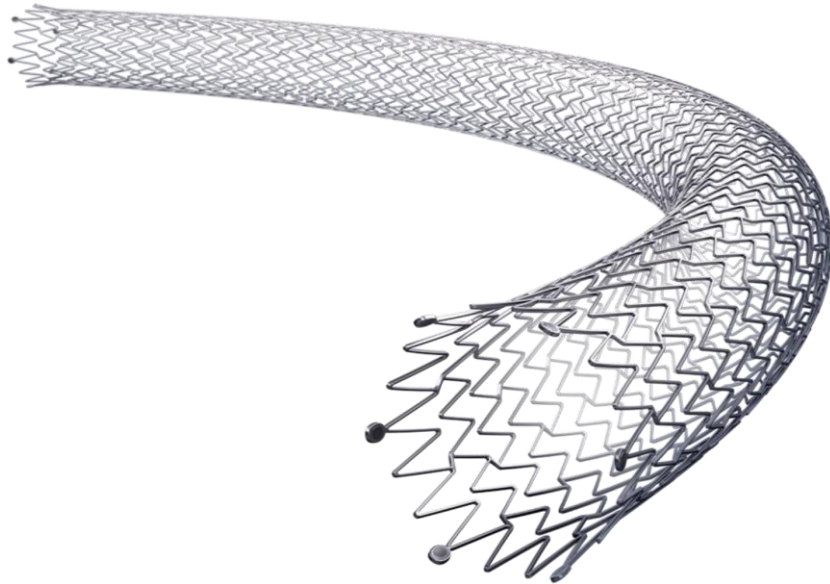
Expanded Polytetrafluoroethylene (ePTFE)



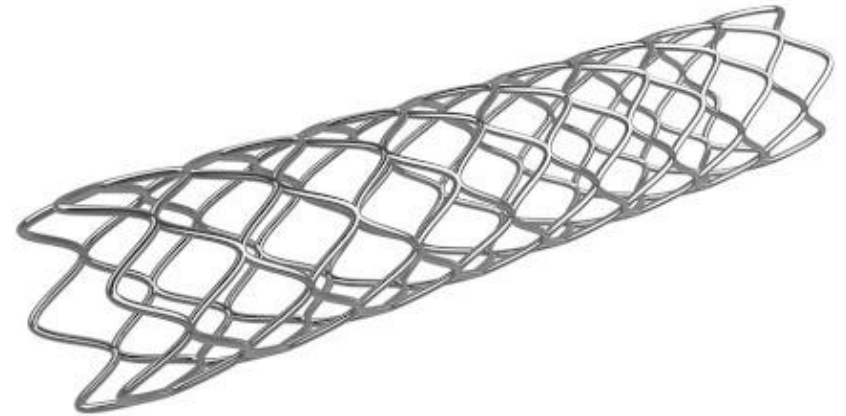
Stent Graft Material Factors. Adapted From <https://evtoday.com/articles/2015-sept-supplement/stent-graft-material-factors-that-impact-limb-complication-rates#:~:text=GRAFT%20MATERIALS,known%20as%20polyester%2FDacron>
Knitted PET Vascular Prosthesis. Adapted From https://www.researchgate.net/figure/A-Knitted-crimped-polyethylene-terephthalate-PET-vascular-prostheses-6-mm-with_fig1_259626204
Invamed Endovascular Stent Graft System. Adapted From <https://invamed.com/product/atlas-endovascular-stent-graft-system-10>
ALTO Abdominal Stent Graft. Adapted From [https://www.jvascsurg.org/article/S0741-5214\(22\)02205-4/fulltext](https://www.jvascsurg.org/article/S0741-5214(22)02205-4/fulltext)

Potential Skeletal Materials

Nitinol (Nickel and Titanium)



Stainless Steel



Current Products

ALTO™
Abdominal Stent
Graft System



1. Adaptive sealing ring
2. Separation of fixation and sealing
3. Integrated balloon system
4. PTFE iliac limbs
5. Helical nitinol mesh

PRO'S

- + Increases Patient Population
- + Integrated Compliant Balloon
- + Separated Fixation/Sealing

CON'S

- Does not address secondary endoleak
- Balloon failure
 - Anaphylaxis

Current Products

GORE® EXCLUDER® AAA ENDOPROSTHESIS



PRO'S

In a 1yr follow up:

no instance of

- + type 1 endoleak
- + type 3 endoleak
- + migration
- + rupture
- + stent fracture
- + limb occlusion

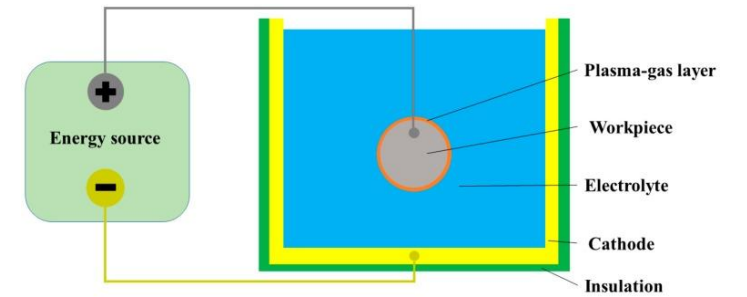
CON'S

- Does not address secondary endoleak

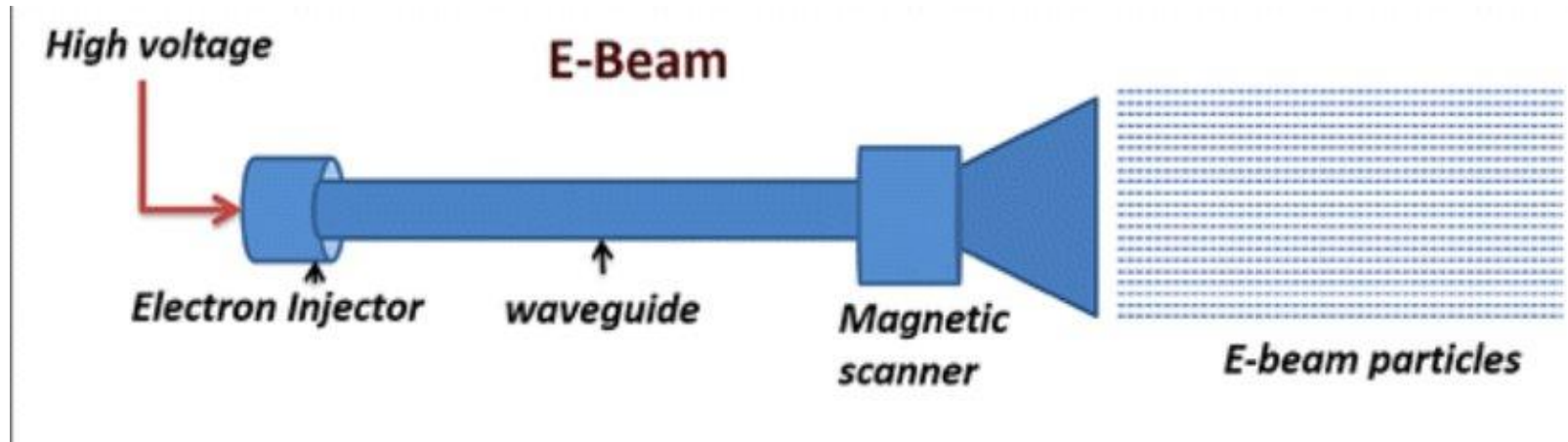
Cleaning Sterilization IP FDA



rbp 



Cleaning



- Internationally Accepted, FDA approved
- High dosage rate/ sterility assurance levels
- Low cost and fast
- Minimal atmospheric effects, no alteration of material properties on stent
- Safe/ negligible emissions

Sterilization

Standards

Management

ISO 13485, 14971, 14155

Sterilization

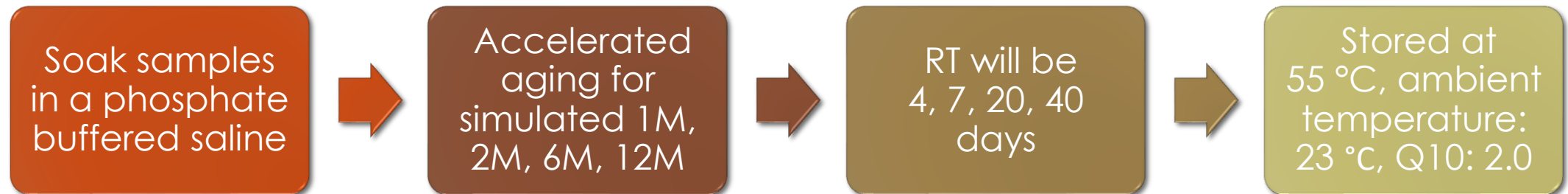
ISO 11137-1

Endovascular Specific

ISO 25539-1, 25539-2, AAMI TIR42:2021



Accelerated Aging



Testing Standards

3-104 ASTM F2914-12 (Reapproved 2018)

Standard Guide for Identification of Shelf-life Test Attributes for Endovascular Devices

3-138 ASTM F2942-13

Standard Guide for the In Vitro Axial, Bending, and Rotational Durability Test of Vascular Stents

3-162 ASTM F3374-19

Standard Guide for Active Fixation Durability of Endovascular Prostheses

3-171 ASTM F2514-21

Standard Guide for Finite Element Analysis (FEA) of Metallic Vascular Stents Subjected to Uniform Radial Loading

3-180 ASTM F3505-21

Standard Test Method for Stent and Endovascular Prosthesis Kink Resistance

3-182 ASTM F3036-21

Standard Guide for Testing Absorbable Stents

3-184 ASTM F2477-23

Standard Test Methods for in vitro Pulsatile Durability Testing of Vascular Stents and Endovascular Prostheses

3-189 ASTM F2942-19

Standard Guide for in vitro Axial, Bending, and Torsional Durability Testing of Vascular Stents

Intellectual Property

- US7407509B2
 - Status: Expired as of 2024-01-13
 - Unique Feature: Added tube of material spiraling down the outside of the trunk to shunt blood
- US9597204B2
 - Status: Active; Expiration Date: 2033-03-10
 - Unique Feature: The trunk of the stent is designed to be twisted and constricted about a midpoint in the trunk to form an hourglass shape
- EP1810642B1
 - Status: Active; Expiration Date: 2027-01-19
 - Unique Feature: Suprarenal fixation to mitigate the possibility of stent migration

FDA

- Class III Medical Device
 - Implant, sustaining life, high risk
- New physical design will require a PMA
 - No current devices in the market address 1a endoleaks in a similar manner
- Testing will follow current ASTM standards.

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